KCE PATHWAYS · GRADE 3 · UNIT 5: DATA AND DESIGN · WEEK 22 · MATH

The Ozark Woodshop — Teacher Lesson Plan & Answer Key

Learning objectives: Students measure lengths to the nearest half inch and quarter inch by lining one end up with 0 (and by counting from the start when an object does not begin at 0), show measurement data on a line plot labeled in halves or quarters with one X for each measurement, read a line plot to find the most common, longest and shortest lengths, and solve two-step length and line plot problems.

Duration: 35–45 minutes
Materials: Computer or tablet with the lesson open · pencil · printed Pre & Post sheet

Lesson flow (as it appears on screen)

1. Title and name: the student types first and last name so results go to KingCretot Experience. Read the Pathway Points rules together: 2 tries per question, points only for a correct first try, and the answer and reason appear after a second miss. Introduce Mr. Calloway's woodworking shop near Mountain View, Arkansas, in the Ozark Mountains, with Clementine, Dashiell, Delia and their cousin Cormac.
2. Warm-Up: Picture Graphs and Bar Graphs Review (10 pts, Week 21 skill): reading a key or scale and how many more or fewer. If a student misses both, review "read the key or the scale first."
3. Picture It: The Ozark Woodshop. The helpers measure every piece for the fall craft fair, and Cormac puts the lengths on a line plot. Ask: why would Mr. Calloway want to see all the sizes at once?
4. Word Wrangler: 6 vocab cards (length, half inch, quarter inch, ruler, line plot, line up with 0).
5. Pencil Break (Pre): the student opens or prints the Pre/Post sheet (grade3/prepost/KCE_Math_PrePost_G3W22.pdf) and completes Page 1 (Pre) before any teaching. Do not help with answers yet; it shows what they already know.
6. Measuring to the Half Inch (10 pts): reading a ruler to the half inch, half-inch differences and a find-the-mistake item about an object that starts at 1. Tip: line up one end with 0, find the last whole inch passed, then check for the middle mark.
7. Measuring to the Quarter Inch (10 pts): the marks after 2 are 2 1/4, 2 1/2, 2 3/4, then 3. Tip: have the student touch and name each mark aloud; 2/4 inch lands on the same mark as 1/2 inch.
8. Making a Line Plot (10 pts): counting how many X's go above a length and choosing the labels. Tip: cross each measurement off the list as its X is drawn.
9. Reading a Line Plot (10 pts): most common (tallest stack), longer than (to the right), shorter than (to the left) and how many more (subtract the counts).
10. Woodshop Challenge: Two-Step Problems (10 pts): adding line plot counts, then comparing or finding how many more; joining half-inch or quarter-inch pieces into whole inches.
11. Bonus Practice: The Sanding Table (XP only, not scored).
12. Match-Up Round (10 pts, 2 per first-try match): Set 1 matches measuring words to meanings, Set 2 matches ruler marks to lengths (2/4 inch = 1/2 inch; 4 quarter inches = 1 inch), Set 3 matches line plot ideas to meanings.
13. Fix the Sentence (10 pts, 5 each): typed answers are checked exactly (capital letters for names of people and places and the title Mr., end mark, commas between a town and a state and in a list, and one verb fix); curly quotes and extra spaces are forgiven. Both serial-comma and no-serial-comma list forms are accepted. Remind the student to keep the same words in the same order.

14. Speed Arena and Keyboard Corral (XP only, not scored).
15. Boss Battle, Part 1: The Craft Fair Table (10 pts, 5 each): counting all the X's, how many more, most common, shorter or longer than, and a find-the-mistake item (most X's is the most common length, not the longest).
16. Boss Battle, Part 2: The Last Cut (10 pts, 5 each): measuring an object that does not start at 0, comparing 3 3/4 and 4, and two-step joining or line plot problems.
17. Pencil Break (Post): the student turns to Page 2 (Post) of the same sheet (grade3/prepost/KCE_Math_PrePost_G3W22.pdf) and completes it, plus the ★ Challenge if time allows, before seeing the score. The sheet buttons let the student open or print it again.
18. Results (Woodshop Day Complete!): the score out of 100 is sent to KingCretot Experience. 85 or higher shows "Woodshop Hero!" with the congratulations message "Woodshop hero! You measured every piece for the craft fair!" and a big confetti burst; below 85 shows "Great Effort Today!" with encouragement ("Every woodworker gets better with practice."), and the next attempt brings a fresh question set (sets rotate 1, 2, 3). Both messages remind the student to finish the Post-Lesson Challenge sheet.

Lesson answer key with reasoning

Every scored item the student sees, in order, across all three rotating question sets. Each item allows 2 tries and earns Pathway Points only on a correct first try. The "why" is the same reasoning shown on screen after a second missed try.

WARM-UP: PICTURE GRAPHS AND BAR GRAPHS REVIEW (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
w1_s0	Set 1: Use the birdhouse picture graph. How many birdhouses were sold on Saturday?	**35 birdhouses** (also accepted: 35)	Each picture stands for 5 birdhouses. Saturday has 7 pictures: 7 × 5 = 35 birdhouses.
w2_s0	Set 1: Use the same graph. How many fewer birdhouses were sold on Friday than on Sunday?	**5 birdhouses** (also accepted: 5 fewer, 5 fewer birdhouses, 5, five, five birdhouses, five fewer, five fewer birdhouses)	Sunday: 5 × 5 = 25. Friday: 4 × 5 = 20. 25 − 20 = 5 fewer. Or: 1 fewer picture × 5 = 5.
w1_s1	Set 2: On a bar graph with a scale of 10, the walnut bowl bar stops halfway between 30 and 40. The oak bowl bar stops on the 60 line. How many more oak bowls than walnut bowls were sold?	**25 bowls** (also accepted: 25 more, 25 more bowls, 25)	Walnut: halfway between 30 and 40 is 35. Oak: 60. 60 − 35 = 25 more oak bowls.
w2_s1	Set 2: Find the mistake. A picture graph key says each spoon picture stands for 2 spoons. Delia says a row with 4 and a half pictures shows 4 and a half spoons. What went wrong? [She forgot the key. 4 pictures × 2 = 8, and half a picture is 1, so the row shows 9 spoons. / Nothing is wrong. 4 and a half pictures means 4 and a half spoons. / The row shows 10 spoons, because the half picture counts as 2. / The row shows 8 spoons, because half pictures do not count.]	**She forgot the key. 4 pictures × 2 = 8, and half a picture is 1, so the row shows 9 spoons.**	Each whole picture is 2 spoons and half a picture is half of 2, which is 1. 8 + 1 = 9 spoons.
w1_s2	Set 3: Use the cutting board bar graph. How many cutting boards were sold on Monday and Thursday together?	**60 boards** (also accepted: 60 cutting boards, 60)	Monday: the bar stops on 25. Thursday: the bar stops on 35. 25 + 35 = 60 cutting boards.

ID	Question	Answer	Why (explain to student)
w2_s2	Set 3: Which explanation is correct? How many more boards were sold on Tuesday than on Wednesday? [Tuesday is 40 and Wednesday is 15, so 40 − 15 = 25 more boards. / Tuesday is 5 grid lines taller, so 5 more boards. / 40 + 15 = 55 more boards. / Tuesday is 40 and Wednesday is 15, so 40 − 15 = 35 more boards.]	**Tuesday is 40 and Wednesday is 15, so 40 − 15 = 25 more boards.**	The scale counts by 5. Tuesday stops on 40 and Wednesday on 15. 40 − 15 = 25. Check: 15 + 25 = 40.

MEASURING TO THE HALF INCH (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
h1_s0	Set 1: How long is this wooden peg? Type your answer like 2 1/2 inches.	**4 1/2 inches** (also accepted, with or without the unit word: 4 and 1/2, four and 1/2, 4 one half, 4 and one half, four and one half, 4 a half, and 73 other spellings of the same answer)	The peg starts at 0, passes the 4, and stops at the middle mark between 4 and 5. It is 4 1/2 inches.
h2_s0	Set 1: Find the mistake. Cormac measured this dowel and says it is 4 1/2 inches long. What went wrong? [The dowel starts at 1, not 0. From 1 to 4 1/2 is 3 1/2 inches. / Nothing is wrong. The dowel ends at 4 1/2, so it is 4 1/2 inches. / It is 5 inches, because you round up. / It is 3 inches, because you only count whole inches.]	**The dowel starts at 1, not 0. From 1 to 4 1/2 is 3 1/2 inches.**	When an object does not start at 0, count the inches from where it starts. 1 to 4 is 3 inches, plus the half inch is 3 1/2 inches.
h1_s1	Set 2: How long is this wooden spoon handle? Type your answer like 3 1/2 inches.	**2 1/2 inches** (also accepted, with or without the unit word: 2 and 1/2, two and 1/2, 2 one half, 2 and one half, two and one half, 2 a half, and 73 other spellings of the same answer)	The handle starts at 0, passes the 2, and stops at the middle mark between 2 and 3. It is 2 1/2 inches.
h2_s1	Set 2: One peg is 2 1/2 inches long. Another peg is 3 inches long. How much longer is the 3-inch peg? Type your answer like 1/2 inch.	**1/2 inch** (also accepted, with or without the unit word: one half, a half, half, ½, and 30 other spellings of the same answer)	From 2 1/2 to 3 is one half-inch jump on the ruler, so the 3-inch peg is 1/2 inch longer.
h1_s2	Set 3: How long is this toy car? Type your answer like 2 1/2 inches.	**5 1/2 inches** (also accepted, with or without the unit word: 5 and 1/2, five and 1/2, 5 one half, 5 and one half, five and one half, 5 a half, and 73 other spellings of the same answer)	The car starts at 0, passes the 5, and stops at the middle mark between 5 and 6. It is 5 1/2 inches.

ID	Question	Answer	Why (explain to student)
h2_s2	Set 3: Which explanation is correct? Why is the middle mark between 3 and 4 called 3 1/2? [The inch from 3 to 4 is split into 2 equal parts. The middle mark is 1/2 inch past 3. / It is the 3rd mark on the ruler, and every third mark is a half. / It is halfway between 0 and 4, so it is half of 4. / You add 3 + 4 and then cut it in half.]	**The inch from 3 to 4 is split into 2 equal parts. The middle mark is 1/2 inch past 3.**	A half inch is 1 of 2 equal parts of an inch. The middle mark of the inch after 3 is 3 and 1/2.

MEASURING TO THE QUARTER INCH (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
q1_s0	Set 1: How long is this wooden wheel axle? Type your answer like 2 3/4 inches.	**3 1/4 inches** (also accepted, with or without the unit word: 3 and 1/4, three and 1/4, 3 one fourth, 3 and one fourth, three and one fourth, 3 a fourth, and 103 other spellings of the same answer)	The axle starts at 0, passes the 3, and stops at the first short mark after 3. That is 1/4 past 3, so it is 3 1/4 inches.
q2_s0	Set 1: Delia's pencil starts at 0 and ends at the third mark after 1 on a quarter-inch ruler. How long is her pencil?	**1 3/4 inches** (also accepted, with or without the unit word: 1 and 3/4, one and 3/4, 1 three fourths, 1 and three fourths, one and three fourths, 1 three quarters, and 73 other spellings of the same answer)	After 1 the marks are 1 1/4, 1 1/2, and 1 3/4. The third mark is 1 3/4 inches.
q1_s1	Set 2: How long is this wooden toy truck? Type your answer like 2 1/4 inches.	**4 3/4 inches** (also accepted, with or without the unit word: 4 and 3/4, four and 3/4, 4 three fourths, 4 and three fourths, four and three fourths, 4 three quarters, and 73 other spellings of the same answer)	The truck starts at 0, passes the 4, and stops at the last short mark before 5. That is 3/4 past 4, so it is 4 3/4 inches.
q2_s1	Set 2: Find the mistake. A peg ends at the first short mark after 2. Dashiell says the peg is 2 1/2 inches long. What went wrong? [The first short mark after 2 is 1/4 inch past 2, so the peg is 2 1/4 inches. 2 1/2 is the middle mark. / Nothing is wrong. Every mark after 2 is a half. / The peg is 3 inches, because it is past the 2. / The peg is 2 3/4 inches, because the first mark is three fourths.]	**The first short mark after 2 is 1/4 inch past 2, so the peg is 2 1/4 inches. 2 1/2 is the middle mark.**	On a quarter-inch ruler the marks after 2 are 2 1/4, 2 1/2, 2 3/4, then 3. The first one is 2 1/4.

ID	Question	Answer	Why (explain to student)
q1_s2	Set 3: How long is this wooden spoon? Type your answer like 2 3/4 inches.	**5 1/4 inches** (also accepted, with or without the unit word: 5 and 1/4, five and 1/4, 5 one fourth, 5 and one fourth, five and one fourth, 5 a fourth, and 103 other spellings of the same answer)	The spoon starts at 0, passes the 5, and stops at the first short mark after 5. It is 5 1/4 inches.
q2_s2	Set 3: Which explanation is correct? Why is 2/4 inch the same length as 1/2 inch? [If an inch is cut into 4 equal parts, 2 of those parts reach the middle mark, the same spot as 1/2 inch. / Because 2 and 4 are both even numbers. / Because 2/4 is bigger than 1/2, so they round to the same. / They are not the same. 2/4 inch is twice as long as 1/2 inch.]	**If an inch is cut into 4 equal parts, 2 of those parts reach the middle mark, the same spot as 1/2 inch.**	2 quarter inches fill half of the inch, so 2/4 inch and 1/2 inch end at the same mark on the ruler.

MAKING A LINE PLOT (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
l1_s0	Set 1: Cormac measured 7 spoon handles: 4 1/4, 4 1/2, 4 1/4, 5, 4 1/2, 4 1/4, and 4 3/4 inches. How many X's go above 4 1/4 on his line plot?	**3 X's** (also accepted: 3 Xs, 3 X, 3, three, three X's, three Xs, three X)	4 1/4 appears 3 times in the list, so it gets 3 X's.
l2_s0	Set 1: Which number line labels should Cormac use for the spoon handle line plot? [4, 4 1/4, 4 1/2, 4 3/4, 5 / 4, 4 1/2, 5 / 1, 2, 3, 4, 5 / 4, 5, 6, 7]	**4, 4 1/4, 4 1/2, 4 3/4, 5**	The handles go from 4 1/4 to 5 inches and were measured to the quarter inch, so the line needs every quarter inch from 4 to 5.
l1_s1	Set 2: Clementine measured 9 craft sticks: 3 1/2, 4, 3 1/2, 4 1/2, 3, 3 1/2, 4, 3 1/2, and 4 1/2 inches. How many X's go above 3 1/2 on her line plot?	**4 X's** (also accepted: 4 Xs, 4 X, 4, four, four X's, four Xs, four X)	3 1/2 appears 4 times in the list, so it gets 4 X's.
l2_s1	Set 2: Find the mistake. Dashiell made Clementine's line plot but drew only one X above 3 1/2, because “it is the same number every time.” What went wrong? [Each measurement gets its own X. Four sticks measured 3 1/2 inches, so it needs 4 X's. / Nothing is wrong. Each length gets only one X. / He should write the number 4 instead of drawing X's. / He should put the X's above 4, because it is the next whole number.]	**Each measurement gets its own X. Four sticks measured 3 1/2 inches, so it needs 4 X's.**	A line plot shows every measurement. One X stands for one stick, so 4 sticks need 4 X's.
l1_s2	Set 3: Use Cormac's line plot. How many pegs did he measure in all?	**10 pegs** (also accepted: 10, ten, ten pegs)	Count every X: 2 + 3 + 4 + 1 = 10 pegs. Each X is one peg.
l2_s2	Set 3: Cormac measures one more peg. It is 1 3/4 inches long. After he adds its X, how many X's will be above 1 3/4?	**5 X's** (also accepted: 5 Xs, 5 X, 5, five, five X's, five Xs, five X)	There are 4 X's above 1 3/4 now. One more peg adds one more X: 4 + 1 = 5.

READING A LINE PLOT (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
r1_s0	Set 1: Use the toy car wheel line plot. Which length is the most common? Type your answer like 2 1/2 inches.	**3 1/2 inches** (also accepted, with or without the unit word: 3 and 1/2, three and 1/2, 3 one half, 3 and one half, three and one half, 3 a half, and 73 other spellings of the same answer)	The tallest stack of X's, 4 X's, is above 3 1/2. So 3 1/2 inches is the most common length.
r2_s0	Set 1: Use the same line plot. How many wheels are longer than 4 inches?	**5 wheels** (also accepted: 5, five, five wheels)	Longer than 4 means to the right of 4: 4 1/2 has 3 X's and 5 has 2 X's. 3 + 2 = 5 wheels.
r1_s1	Set 2: Use the wooden spoon line plot. How many more spoons are 5 1/2 inches long than 6 inches long?	**3 spoons** (also accepted: 3 more, 3 more spoons, 3, three, three spoons, three more, three more spoons)	5 1/2 has 4 X's and 6 has 1 X. 4 − 1 = 3 more spoons.
r2_s1	Set 2: Find the mistake. Clementine says 6 inches is the most common spoon length, because 6 is in the middle of the line plot. What went wrong? [Most common means the length with the most X's. 5 1/2 has 4 X's, the most, so 5 1/2 inches is the most common. / Nothing is wrong. The middle length is always the most common. / The most common length is 7 inches, because it is the longest. / The most common length is 5 inches, because it is first.]	**Most common means the length with the most X's. 5 1/2 has 4 X's, the most, so 5 1/2 inches is the most common.**	Look for the tallest stack of X's, not the middle of the line. 5 1/2 has 4 X's; 6 has only 1.
r1_s2	Set 3: Use the drawer knob line plot. How many knobs are 2 1/2 inches or shorter?	**7 knobs** (also accepted: 7, seven, seven knobs)	2 1/2 or shorter means 2, 2 1/4, and 2 1/2: 3 + 0 + 4 = 7 knobs.
r2_s2	Set 3: Use the same line plot. How much longer is the longest knob than the shortest knob? Type your answer like 2 inches.	**1 inch** (also accepted: one inch, 1, one, 1 inches, 1 in, 1 in., one inches, one in, one in.)	The shortest knob is 2 inches and the longest is 3 inches. From 2 to 3 is 1 inch.

WOODSHOP CHALLENGE: TWO-STEP PROBLEMS (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
t1_s0	Set 1: Two steps! Delia's line plot shows 4 pegs at 2 1/4 inches, 6 pegs at 2 1/2 inches, and 3 pegs at 2 3/4 inches. A birdhouse kit needs 15 pegs. How many more pegs must she cut?	**2 pegs** (also accepted: 2 more pegs, 2, two, two pegs, two more pegs)	Step 1: 4 + 6 + 3 = 13 pegs. Step 2: 15 − 13 = 2 more pegs.
t2_s0	Set 1: Mr. Calloway glues two pieces of trim that are each 1/2 inch long end to end. Then he adds a 3-inch piece. How long is the whole strip? Type your answer like 5 inches.	**4 inches** (also accepted: four inches, 4, four, 4 inch, 4 in, 4 in., four inch, four in, four in.)	Step 1: 1/2 inch + 1/2 inch = 1 whole inch. Step 2: 1 + 3 = 4 inches.
t1_s1	Set 2: Two steps! Dashiell's line plot shows 5 spoons at 6 inches, 2 spoons at 6 1/2 inches, and 9 spoons at 7 inches. How many more 7-inch spoons are there than 6-inch and 6 1/2 -inch spoons together?	**2 spoons** (also accepted: 2 more, 2 more spoons, 2, two, two spoons, two more, two more spoons)	Step 1: 5 + 2 = 7 shorter spoons. Step 2: 9 − 7 = 2 more 7-inch spoons.

ID	Question	Answer	Why (explain to student)
t2_s1	Set 2: Clementine strings 4 beads that are each 1/4 inch long, and then a 2-inch bead. How long is the row of beads? Type your answer like 5 inches.	**3 inches** (also accepted: three inches, 3, three, 3 inch, 3 in, 3 in., three inch, three in, three in.)	Step 1: 4 quarter inches make 1 whole inch. Step 2: 1 + 2 = 3 inches.
t1_s2	Set 3: Which explanation is correct? Cormac's line plot has 3 X's above 4 1/2 and 5 X's above 5. Then he cuts 2 more pieces that are 4 1/2 inches long. Which length has more X's now? [Neither. 3 + 2 = 5 X's above 4 1/2, and there are 5 X's above 5, so they are equal. / 4 1/2, because he added X's to it. / 5, because 5 is a bigger number than 4 1/2. / 4 1/2, because it now has 2 X's.]	**Neither. 3 + 2 = 5 X's above 4 1/2, and there are 5 X's above 5, so they are equal.**	Step 1: 3 + 2 = 5 X's above 4 1/2. Step 2: compare 5 and 5. They are the same.
t2_s2	Set 3: Each table leg piece must be at least 3 1/2 inches long. The line plot shows 2 pieces at 3 1/4, 4 pieces at 3 1/2, 3 pieces at 3 3/4, and 1 piece at 4 inches. How many pieces are long enough?	**8 pieces** (also accepted: 8, eight, eight pieces)	At least 3 1/2 means 3 1/2 or longer: 4 + 3 + 1 = 8 pieces. The 3 1/4 pieces are too short.

MATCH-UP ROUND (2 POINTS PER FIRST-TRY MATCH)

ID	Question	Answer	Why (explain to student)
match_s0_1	Set 1: Match half inch	**1 of 2 equal parts of an inch**	2 points if matched on the first try.
match_s0_2	Set 1: Match quarter inch	**1 of 4 equal parts of an inch**	2 points if matched on the first try.
match_s0_3	Set 1: Match line plot	**a number line with an X for each measurement**	2 points if matched on the first try.
match_s0_4	Set 1: Match ruler	**a tool for measuring length**	2 points if matched on the first try.
match_s0_5	Set 1: Match one X	**one measurement**	2 points if matched on the first try.
match_s1_1	Set 2: Match first short mark after 3	**3¼ inches**	2 points if matched on the first try.
match_s1_2	Set 2: Match middle mark between 3 and 4	**3½ inches**	2 points if matched on the first try.
match_s1_3	Set 2: Match mark just before 4	**3¾ inches**	2 points if matched on the first try.
match_s1_4	Set 2: Match 2/4 inch	**the same as 1/2 inch**	2 points if matched on the first try.
match_s1_5	Set 2: Match 4 quarter inches	**1 whole inch**	2 points if matched on the first try.
match_s2_1	Set 3: Match most common length	**the tallest stack of X's**	2 points if matched on the first try.
match_s2_2	Set 3: Match count all the X's	**how many things were measured**	2 points if matched on the first try.
match_s2_3	Set 3: Match no X's above a length	**nothing measured that length**	2 points if matched on the first try.
match_s2_4	Set 3: Match start measuring	**line up one end with 0**	2 points if matched on the first try.
match_s2_5	Set 3: Match 1/2 inch + 1/2 inch	**1 inch**	2 points if matched on the first try.

FIX THE SENTENCE (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
fs1_s0	Set 1: Fix the sentence: mr calloway works in mountain view arkansas	**Mr. Calloway works in Mountain View, Arkansas.**	Mr. is a title, so it gets a capital M and a period. Calloway, Mountain View, and Arkansas are names, so they get capital letters. Put a comma between the town and the state, and end with a period.
fs2_s0	Set 1: Fix the sentence: we measured pegs spoons and wheels	**We measured pegs, spoons, and wheels.** (also accepted: We measured pegs, spoons and wheels.)	Start with a capital W, put commas between the things in the list, and end with a period.
fs1_s1	Set 2: Fix the sentence: how long is the walnut spoon	**How long is the walnut spoon?**	Start with a capital H. The sentence asks something, so it ends with a question mark.
fs2_s1	Set 2: Fix the sentence: clementine dashiell and delia sanded the boards	**Clementine, Dashiell, and Delia sanded the boards.** (also accepted: Clementine, Dashiell and Delia sanded the boards.)	Each name gets a capital letter, commas go between the names in the list, and the sentence ends with a period.
fs1_s2	Set 3: Fix the sentence: the ozark mountains are in arkansas	**The Ozark Mountains are in Arkansas.**	Start with a capital T. Ozark Mountains and Arkansas are names of places, so each word gets a capital letter. End with a period.
fs2_s2	Set 3: Fix the sentence: the four pegs was the same length	**The four pegs were the same length.**	Start with a capital T and end with a period. Four pegs means more than one, so use were, not was.

BOSS BATTLE: THE CRAFT FAIR TABLE (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
bb1_s0	Set 1: Use the toy car line plot. How many toy cars were measured?	**12 cars** (also accepted: 12 toy cars, 12)	Count every X: 2 + 1 + 4 + 3 + 2 = 12 toy cars.
bb2_s0	Set 1: Use the same line plot. How many more cars are 3 1/2 inches long than 3 1/4 inches long?	**3 cars** (also accepted: 3 more, 3 more cars, 3, three, three cars, three more, three more cars)	3 1/2 has 4 X's and 3 1/4 has 1 X. 4 − 1 = 3 more cars.
bb1_s1	Set 2: Use the shelf piece line plot. Which length is the most common? Type your answer like 2 1/2 inches.	**3 1/2 inches** (also accepted, with or without the unit word: 3 and 1/2, three and 1/2, 3 one half, 3 and one half, three and one half, 3 a half, and 73 other spellings of the same answer)	The tallest stack, 5 X's, is above 3 1/2. So 3 1/2 inches is the most common length.
bb2_s1	Set 2: Use the same line plot. How many pieces are shorter than 3 inches?	**3 pieces** (also accepted: 3, three, three pieces)	Shorter than 3 means to the left of 3. Only 2 1/2 is there, with 3 X's, so 3 pieces.
bb1_s2	Set 3: Use the whistle piece line plot. How many pieces are longer than 1 1/2 inches?	**6 pieces** (also accepted: 6, six, six pieces)	Longer than 1 1/2 means 1 3/4 and 2: 4 + 2 = 6 pieces.

ID	Question	Answer	Why (explain to student)
bb2_s2	Set 3: Find the mistake. Dashiell says the longest whistle piece is 1 3/4 inches, because 1 3/4 has the most X's. What went wrong? [The most X's tells the most common length, not the longest. The X farthest to the right is above 2, so the longest piece is 2 inches. / Nothing is wrong. The length with the most X's is always the longest. / The longest piece is 1 inch, because it is first on the line. / The longest piece is 1 1/4 inches, because it has 3 X's.]	**The most X's tells the most common length, not the longest. The X farthest to the right is above 2, so the longest piece is 2 inches.**	Longest means the biggest length that has an X. 2 inches has 2 X's and is farthest right.

BOSS BATTLE: THE LAST CUT (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
bb3_s0	Set 1: This toy car's back end is lined up with the 1, not the 0. How long is the car? Type your answer like 2 1/4 inches.	**3 3/4 inches** (also accepted, with or without the unit word: 3 and 3/4, three and 3/4, 3 three fourths, 3 and three fourths, three and three fourths, 3 three quarters, and 73 other spellings of the same answer)	The car goes from 1 to 4 3/4. From 1 to 4 is 3 inches, and then 3/4 more: 3 3/4 inches.
bb4_s0	Set 1: Find the mistake. Delia says 3 3/4 inches is longer than 4 inches, because three fourths is a big part. What went wrong? [3 3/4 is only 3/4 inch past 3, and 4 is a whole inch past 3. So 3 3/4 is shorter than 4. / Nothing is wrong. 3 3/4 is longer than 4. / They are the same length, because both are close to 4. / 3 3/4 is longer, because it has more numbers in it.]	**3 3/4 is only 3/4 inch past 3, and 4 is a whole inch past 3. So 3 3/4 is shorter than 4.**	On a ruler, 3 3/4 is the mark just before 4. It is 1/4 inch shorter than 4 inches.
bb3_s1	Set 2: How long is this shelf bracket? Type your answer like 2 1/4 inches.	**5 3/4 inches** (also accepted, with or without the unit word: 5 and 3/4, five and 3/4, 5 three fourths, 5 and three fourths, five and three fourths, 5 three quarters, and 73 other spellings of the same answer)	The bracket starts at 0, passes the 5, and stops at the last short mark before 6: 5 3/4 inches.
bb4_s1	Set 2: Mr. Calloway needs 6 shelf pieces that are 4 inches or longer. Using the shelf piece line plot, how many more of those pieces must he cut?	**3 pieces** (also accepted: 3 more, 3 more pieces, 3, three, three pieces, three more, three more pieces)	Step 1: 4 inches has 1 X and 4 1/2 has 2 X's, so 1 + 2 = 3 pieces are 4 inches or longer. Step 2: 6 − 3 = 3 more pieces.
bb3_s2	Set 3: This whistle's left end is lined up with the 2, not the 0. How long is the whistle? Type your answer like 2 1/2 inches.	**3 1/4 inches** (also accepted, with or without the unit word: 3 and 1/4, three and 1/4, 3 one fourth, 3 and one fourth, three and one fourth, 3 a fourth, and 103 other spellings of the same answer)	The whistle goes from 2 to 5 1/4. From 2 to 5 is 3 inches, and then 1/4 more: 3 1/4 inches.

ID	Question	Answer	Why (explain to student)
bb4_s2	Set 3: Cormac glues two blocks that are each 1/2 inch long onto the end of a 2 1/4 -inch block. How long is the new piece? Type your answer like 4 1/2 inches.	**3 1/4 inches** (also accepted, with or without the unit word: 3 and 1/4, three and 1/4, 3 one fourth, 3 and one fourth, three and one fourth, 3 a fourth, and 103 other spellings of the same answer)	Step 1: 1/2 inch + 1/2 inch = 1 inch. Step 2: 2 1/4 + 1 = 3 1/4 inches.

Pre & Post sheet answer guide

Many items are open-ended. Grade for genuine understanding and complete sentences, not exact wording. Sheet: grade3/prepost/KCE_Math_PrePost_G3W22.pdf.

PRE-LESSON WARM-UP

#	Prompt	Model answer / grading note
1	Review: key of 5; a row of 6 pictures vs. a row of 4	6 × 5 = 30 birdhouses. A row of 4 pictures is 20, so 30 − 20 = 10 more.
2	Draw a line exactly 3 inches long	The line starts at 0 and ends on the 3. Check with a ruler.
3	Equal parts in each inch at the half-inch marks; at the quarter-inch marks	2 equal parts at the half-inch marks; 4 equal parts at the quarter-inch marks.
4	Which is longer: 2 1/2 inches or 2 inches?	2 1/2 inches
5	Measure something small at home to the nearest half inch	Answers vary. Check that the student names the object, starts at 0, and writes a length in whole or half inches.

POST-LESSON CHALLENGE

#	Prompt	Model answer / grading note
1	Draw a 4 1/2-inch line and a 2 3/4-inch line	Check with a ruler: each line starts at 0; one ends at the middle mark between 4 and 5, the other at the third quarter-inch mark after 2.
2	Line plot of 8 pegs (labels 2, 2 1/4, 2 1/2, 2 3/4, 3)	X's: 2 → 0, 2 1/4 → 4, 2 1/2 → 2, 2 3/4 → 1, 3 → 1. That is 8 X's in all.
3	Most common length; pegs longer than 2 1/2 inches	2 1/4 inches is most common (4 X's). 2 pegs are longer than 2 1/2 inches (one at 2 3/4 and one at 3).
4	Delia's spoon starts at 1 and ends at 5 1/2	She did not start at 0, so she read the end mark instead of counting from 1. From 1 to 5 1/2 is 4 1/2 inches.

★ CHALLENGE

#	Prompt	Model answer / grading note
1	Cormac needs 10 pegs that are 2 1/2 inches or longer	6 more pegs. The line plot has 2 + 1 + 1 = 4 pegs at 2 1/2 inches or longer, and 10 − 4 = 6.